

Koushik Kolla

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SUMMARY

Information Science Engineering student with hands-on experience in Enterprise Java (Java 21, Spring Boot, Spring Batch), enterprise integrations (Apache Camel), and microservice containerization. Experienced in designing robust REST/SOAP web services, type-safe database queries with jOOQ, and building scalable full-stack web applications.

EDUCATION

Global Academy of Technology (Affiliated to VTU)

Dec 2022 – Jul 2026

Bachelor of Engineering in Information Science and Engineering

GPA: 8.39 / 10.0

TECHNICAL SKILLS

Languages: Java (Java 21), SQL, Python, JavaScript, HTML/CSS

Frameworks & Backend: Spring Boot, Spring Batch, Apache Camel, Node.js, Express.js, React.js

Data Access & Databases: jOOQ, MySQL, MongoDB, JDBC

Web Services & Testing: REST APIs, SOAP, JUnit, ReadyAPI, Bruno, Postman

DevOps & Cloud: Docker, Kubernetes, Maven, Google Cloud (Vertex AI)

Tools & Environments: Git, GitHub, Linux/Bash, WinSCP, PuTTY, VS Code

EXPERIENCE

Tieto

Apr, 2026 – Oct, 2026

Software Engineering Apprentice

Bengaluru, India

- Engineered backend microservices utilizing **Java 21** and **Spring Boot**, adhering to enterprise design patterns, clean coding standards, and modular architecture.
- Developed robust data pipelines and asynchronous workflows using **Spring Batch** and **Apache Camel** for automated message routing and bulk data processing.
- Designed type-safe relational database operations using **jOOQ** and **SQL**, improving query execution performance and eliminating runtime syntax vulnerabilities.
- Built and documented secure **RESTful** and **SOAP** web services; authored automated unit and integration tests using **JUnit** to ensure high code coverage.
- Tested API contracts using **Bruno** and **ReadyAPI**; containerized applications with **Docker** and managed remote server configurations via **PuTTY** and **WinSCP**.

PROJECTS

Gesture-Based E-Voting System & Facial Authentication | *Python, OpenCV, MediaPipe, SQL*

- Engineered a contactless biometric authentication pipeline using OpenCV, achieving **95%+ face verification accuracy** and reducing identity verification time to **under 1.5s**.
- Implemented real-time hand-tracking with MediaPipe at **30+ FPS**, incorporating a **2-second hold validation** algorithm to eliminate accidental ballot submissions.
- Designed role-based access control (Admin, Candidate, Voter) over a relational database, enforcing single-vote integrity constraints to prevent **100% of duplicate ballots**.

Personalized News Digest Service | *React, Node.js, Express, MongoDB, Tailwind CSS*

- Architected a full-stack automated news pipeline, scheduling daily cron jobs to fetch, filter, and dispatch batch digests via Nodemailer with **99.5% delivery reliability**.
- Optimized MongoDB performance by indexing user interest schemas and deduplicating NewsAPI responses, cutting payload size and reducing read latency by **~35%**.
- Developed a responsive dashboard in React and Tailwind featuring JWT authentication and dynamic preference toggles, achieving **sub-1.2s page load times**.

Enterprise Bank Account Management System | *Java 21, JDBC, MySQL, OOP*

- Engineered a multi-tiered banking engine in Java to manage customer onboarding, balance inquiries, and fund transfers using Object-Oriented principles and modular design.
- Integrated MySQL using JDBC, implementing ACID-compliant transaction rollback mechanisms to maintain zero data inconsistencies during concurrent account transfers.
- Built robust validation layers and custom exception handling architectures to simulate enterprise-grade financial transaction flows securely.